
MIDTERM LAB EXAM

Introduction to Programming

CSE141

Instructor: Dr Ahmed Akhtar

Date: 9th October, 2026

Duration: 2 hours

Total Marks: 90

INSTRUCTIONS

1. There are a total of 4 questions out of which you must **attempt any 3 questions** carrying **30 marks** each.
2. Submitting only the **.out** or **.exe** file for a question will result in **0 marks**.
3. You will be given 5 minutes at the end to submit the files on **LMS**.
4. Only **VS Code** should be open throughout the duration of the exam. Before the specified time, you are not allowed to open the **web browser**.
5. Not following the instructions will result in **negative marking**.

Do not turn this page until instructed.

Question 1**[30 marks]**

Write a program that reads a string and an integer N representing the number of lines. The program should distribute the characters of the input string across the specified number of lines in a round-robin pattern. Each character of the input string is placed on a line based on its position in the sequence (the first character goes to line 1, the second to line 2, and so on, wrapping around after the last line), with the characters on each line printed and separated by spaces.

Sample Input:

```
Enter a string : Hello World
Enter number of lines : 3
```

Sample Output:

```
H   l   W   l
 e   o   o   d
  l       r
```

Question 2**[30 marks]**

Write a program that implements matrix multiplication using one-dimensional arrays (or `std::vector`) instead of two-dimensional ones. The program should read two matrices of integers from the user: the first matrix A of size $N \times K$ and the second matrix B of size $K \times M$. Store both matrices as 1D arrays, then multiply them to produce a resulting matrix C of size $N \times M$, also represented as a 1D array. Each element of the result is computed as the dot product of a row from A and a column from B , according to the formula:

$$C_{ij} = \sum_{k=1}^K A_{ik} \times B_{kj}$$

Since the matrices are stored in one-dimensional arrays, use index mapping to access elements

$$A[i][j] \rightarrow A[i \times K + j] \quad (1)$$

$$B[i][j] \rightarrow B[i \times M + j] \quad (2)$$

$$C[i][j] \rightarrow C[i \times M + j] \quad (3)$$

After computing the result, print the resulting matrix in a formatted $N \times M$ layout, where each row is displayed on a separate line.

Sample Input:

```
Enter dimensions N, K, M: 3 4 5
Enter matrix A (3x4):
1 2 3
4 5 6
7 8 9
1 2 3
Enter matrix B (4x5):
```

```
1 0 0 0
0 0 1 0
0 0 0 1
0 1 0 0
0 0 0 0
```

Sample Output:

Resulting matrix C (3x5):

```
1 5 0 3 0
5 13 0 7 0
9 3 0 2 0
```

Question 3**[30 marks]**

Write a program that opens two files, `file1.txt` and `file2.txt`. Read them simultaneously and write their lines alternately into a new file `merged.txt` (Line 1 from file 1, Line 1 from file 2, Line 2 from file 1, Line 2 from file 2, etc.). If one file runs out of lines before the other, simply append the remaining lines of the longer file to the end.

Sample Input (`file1.txt`):

```
Assembly
C++
```

Sample Input (`file2.txt`):

```
JavaScript
Go
Rust
```

Sample Output (`merged.txt`):

```
Assembly
JavaScript
C++
Go
Rust
```

Question 4**[30 marks]**

Write a program `max_ones.cpp` that, given an array of integers, finds the maximum number of consecutive 1's present in the array.

Sample Input:

Enter size of array: 9

Enter elements: 1 1 0 1 1 1 0 1 1

Sample Output:

Maximum consecutive 1s: 3

END OF EXAM

Make sure you have uploaded .cpp files of all your attempted questions to LMS Lab Exam submission tab and pressed the final Submit button before you leave.
